

1 Actuator Sizing

1.1 Actuator Sizing for Gripper



Figure 1: Gripper Mechanism

1- Normal Force Move Up:

$$F_{up} \geq \frac{m(g + a)}{2u}$$
$$F_{up} = \frac{m(g + a)}{2u} \times S.f$$

The required force (F) is calculated as:

$$F_{up} = \frac{0.04 \times (9.8 + .4)}{2 \times 0.4} \times 2$$
$$F_{up} = 1.02 \text{ N}$$

Calculation of Torque:

$$T = F \times d$$
$$T_{up} = 1.02 \times .1 = 0.102 \text{ N.m}$$

2- Force Move Sideways:

$$F_{req} = \sqrt{fx^2 + fy^2}$$
$$F_{req} = \sqrt{(ma)^2 + (mg)^2}$$

The required force (F) is calculated as:

$$F_{side} = \frac{\sqrt{(ma)^2 + (mg)^2}}{2u} \times S.f$$
$$F_{side} = \frac{\sqrt{(.04 \times .4)^2 + (9.8 \times .04)^2}}{2 \times .4} \times 2$$
$$F_{side} = .98 \text{ N}$$

Calculation of Torque:

$$T = F \times d$$
$$T_{side} = .98 \times .1 = 0.098 \text{ N.m}$$

Symbol	Value	Units
u_s	0.4	
g	9.8	m/s ²
$S.f$	2	
m	0.04	kg
a	0.4	m/s ²
d	0.1	m

3- Force Move down:

$$F_{down} \geq \frac{m(g - a)}{2u}$$

$$F_{down} = \frac{m(g - a)}{2u} \times S.f$$

The required force (F) is calculated as:

$$F_{down} = \frac{0.04 \times (9.8 - .4)}{2 \times 0.4} \times 2$$

$$F_{down} = .94 \text{ N}$$

Calculation of Torque:

$$T = F \times d$$

$$T_{down} = .94 \times .1 = 0.094 \text{ N.m}$$

4- Validation: From analysis, the gripper mechanism requires a peak torque = 0.102 N-m. The selected MG996R Servo Motor has a maximum torque of 1.1 N-m (in its datasheet). Since this value is significantly higher than the required peak torque = 0.102 N-m, the motor provides a large safety margin and ensures stable and reliable gripping performance.

1.2 Actuator Sizing for Z-Axis

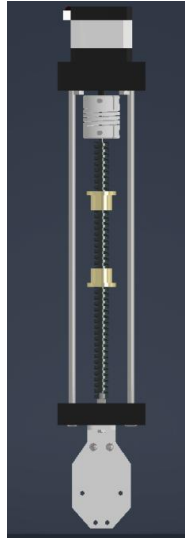


Figure 2: Z-Axis Mechanism

Symbol	Value Used	Units (SI)
m_L	1.118	kg
η	0.40	
ρ	7800	kg/m ³
D	8	mm
L	0.400	m
P	0.125	rev/mm
g	9.81	m/s ²
t_{acc}	0.160	s
t_c	0.500	s
t_{dec}	0.160	s
θ_{acc}	392.70	rad/s ²
Θ'_{max}	62.83	rad/s
$N_{inertia}$	1	$J_{L,total}/J_m$
$S.f$	1.5	

1- Inertia Calculations

Effective Load Inertia:

$$J_L = \frac{m_L}{\eta[2\pi p]^2}$$

$$J_L = \frac{1.118}{.40[2\pi \times 125]^2}$$

$$J_L = 4.5302 \times 10^{-6} \text{ kg.m}^2$$

Lead Screw Inertia:

$$J_{screw} = \frac{\pi}{2} \rho L r^4$$

$$J_{screw} = \frac{\pi}{2} \times 7800 \times .4 \times .004^4$$

$$J_{screw} = 1.2546 \times 10^{-6} \text{ kg.m}^2$$

Total Load Inertia:

$$J_{Lt} = J_{screw} + J_L = 5.7848 \times 10^{-6} \text{ kg.m}^2$$

Equivalent Inertia:

$$J_{Lt} = J_m$$

$$J_{eq} = J_{Lt} + J_m = 2J_{Lt}$$

$$J_{eq} = 1.1570 \times 10^{-5} \text{ kg.m}^2$$

2- Load Torque Calculation

Load Force:

$$f_L = M_L \times g$$

$$f_L = 1.118 \times 9.8 = 10.9694 \text{ N}$$

Load Torque:

$$T_L = \frac{F_L}{2\pi p \times \eta}$$

$$T_L = 0.0349 \text{ N.m}$$

3- Motion Profile

Assume a motion profile where:

$$\Theta'_{max} = 62.83 \text{ rad.s}^{-1}, t_{acc} = 0.160 \text{ s}, t_c = 0.500 \text{ s}, t_{dec} = 0.160 \text{ s}, t_{dwell} = 0.$$

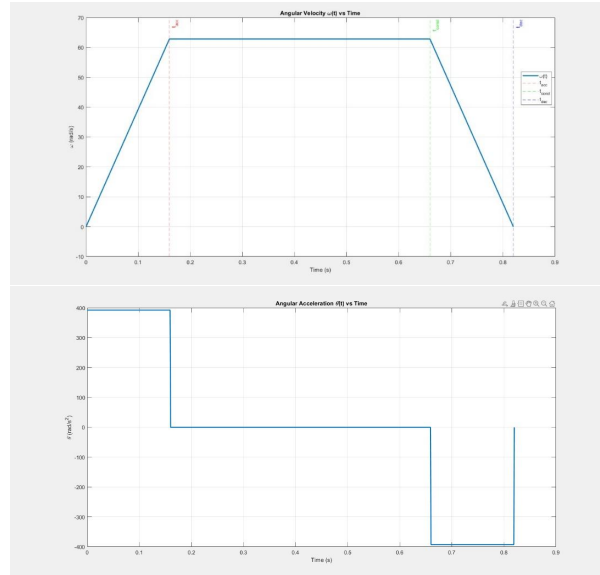


Figure 3: Angular Velocity and Acceleration vs Time

Now calculate the angular acceleration:

$$\theta_{acc} = 392.70 \text{ rad.s}^{-2}$$

$$\theta_{const} = 0$$

$$\theta_{dec} = -392.70 \text{ rad.s}^{-2}$$

4- Peak and RMS Torque Calculations

Acceleration Torque:

$$Tm = J_{eq}\ddot{\theta} + T_L$$

$$Tm_{acc} = (1.1570 \times 10^{-5} \times 392.7) + 0.0349$$

$$Tm_{acc} = 0.0395 \text{ N.m}$$

$$T_{peak} = Tm_{acc} \times S.f = 0.05925 \text{ N.m}$$

Constant Velocity Torque:

$$Tm_c = T_L = 0.0349 \text{ N.m}$$

Deceleration Torque:

$$Tm_{dec} = (1.1570 \times 10^{-5} \times -392.7) + 0.0349$$

$$Tm_{dec} = 0.0304 \text{ N.m}$$

Peak Torque (RMS):

$$T_{rms} = \sqrt{\frac{Tm_{acc}^2 \times t_{acc} + Tm_c^2 \times t_c + Tm_{dec}^2 \times t_{dec}}{t_{acc} + t_c + t_{dec}}} \times S.f$$

$$T_{rms} = \sqrt{\frac{0.0395^2 \times 0.16 + 0.0349^2 \times 0.5 + 0.0304^2 \times 0.16}{.820}} \times 1.5$$

$$T_{rms} = 0.05265 \text{ N.m}$$

5- Validation Based on the analysis, the Z-axis linear motion requires a peak torque = 0.05925 N-m and an RMS torque = 0.05265 N-m. The selected NEMA 17 Stepper Motor has a maximum torque = 0.59 N-m. Since this value is much higher, the motor provides a large safety margin.

1.3 Actuator Sizing for Y-Axis

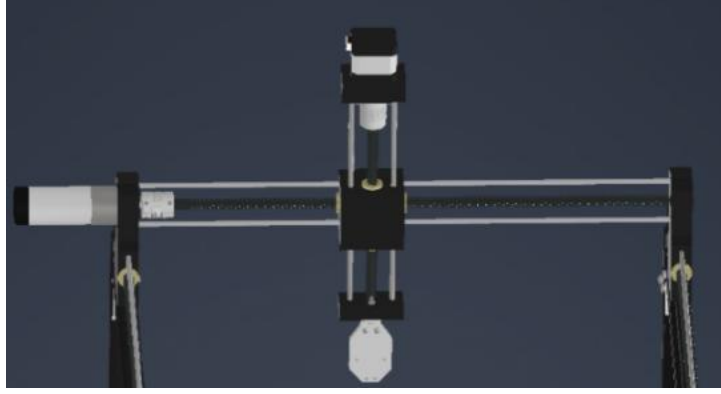


Figure 4: Y-Axis Mechanism

Symbol	Value	Units
η	0.40	
P	0.125	rev/mm
D	8.0	mm
m_L	1.320	kg
L_{screw}	600	mm
$S.f$	1.5	
g	9.81	m/s ²
t_{acc}	0.1600	s
t_c	4.8400	s
t_{dec}	0.1600	s
Θ'_{max}	62.8319	rad/s
θ_{acc}	392.6991	rad/s ²
$N_{inertia}$	1.0	

1- Inertia Calculations

Effective Load Inertia:

$$J_L = \frac{1.320}{.40[2\pi \times 125]^2} = 5.349 \times 10^{-6} \text{ kg.m}^2$$

Lead Screw Inertia:

$$J_{screw} = \frac{\pi}{2} \times 7800 \times .6 \times .004^4 = 1.882 \times 10^{-6} \text{ kg.m}^2$$

Total Load Inertia:

$$J_{Lt} = J_{screw} + J_L = 7.2304 \times 10^{-6} \text{ kg.m}^2$$

Equivalent Inertia:

$$J_{eq} = 2J_{Lt} = 1.4461 \times 10^{-5} \text{ kg.m}^2$$

2- Load Torque Calculation

Load Force:

$$f_L = 1.320 \times 9.8 = 12.936 \text{ N}$$

Load Torque:

$$T_L = \frac{F_L}{2\pi p \times \eta} = 0.0412 \text{ N.m}$$

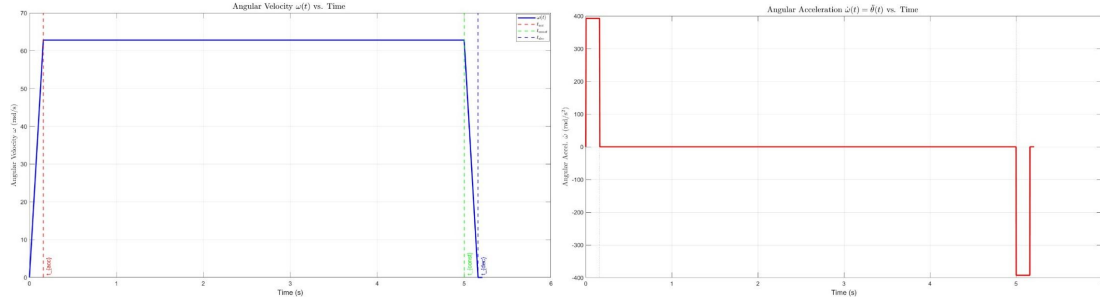


Figure 5: Angular Velocity and Acceleration vs Time (Y-Axis)

4- Peak and RMS Torque Calculations

$$Tm_{acc} = (1.4461 \times 10^{-5} \times 392.7) + 0.0412 = 0.0469 \text{ N.m}$$

$$T_{peak} = Tm_{acc} \times S.f = .0703 \text{ N.m}$$

Deceleration Torque:

$$Tm_{dec} = (1.4461 \times 10^{-5} \times -392.7) + 0.0412 = 0.0355 \text{ N.m}$$

$$T_{rms} = \sqrt{\frac{0.0469^2 \times 0.16 + 0.0412^2 \times 4.8400 + 0.0355^2 \times 0.16}{5.16}} \times 1.5 = 0.0618 \text{ N.m}$$

5- Validation Based on the analysis, the Y-axis linear motion requires a peak torque = 0.0703 N-m and an RMS torque = 0.0618 N-m. The selected 25GA370 DC Gear Motor with Magnetic Encoder has a stall torque = 0.765 N-m. Since this value is much higher, the motor provides a large safety margin.

1.4 Actuator Sizing for X-Axis



Figure 6: X-Axis Mechanism

Symbol	Value	Units
m_L	2.200	kg
L_{screw}	500	mm
t_{acc}	0.1600	s
t_c	5.465	s

Effective Load Inertia:

$$J_L = \frac{2.200}{.40[2\pi \times 125]^2} = 8.916 \times 10^{-6} \text{ kg.m}^2$$

Total Load Inertia:

$$J_{Lt} = J_{screw} + J_L = 1.0484 \times 10^{-5} \text{ kg.m}^2$$

Equivalent Inertia:

$$J_{eq} = 2J_{Lt} = 2.0969 \times 10^{-5} \text{ kg.m}^2$$

Load Force:

$$f_L = 2.2 \times 9.8 = 21.56 \text{ N}$$

$$T_L = \frac{F_L}{2\pi p \times \eta} = 0.0686 \text{ N.m}$$

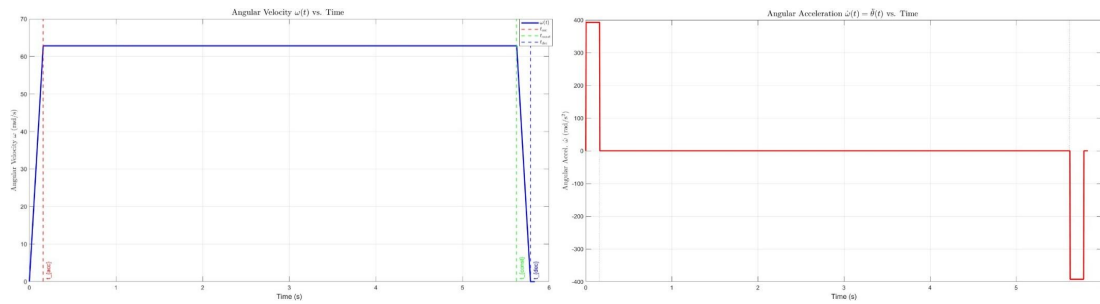


Figure 7: Angular Velocity and Acceleration vs Time (X-Axis)

$$Tm_{acc} = (2.0969 \times 10^{-5} \times 392.7) + 0.0686 = 0.0769 \text{ N.m}$$

$$T_{peak} = Tm_{acc} \times S.f = .115 \text{ N.m}$$

$$T_{rms} = \sqrt{\frac{0.0769^2 \times 0.16 + 0.0686^2 \times 5.465 + 0.0604^2 \times 0.16}{5.785}} \times 1.5 = 0.103 \text{ N.m}$$

5- Validation

Based on the analysis, the X-axis linear slide requires a peak torque = 0.115 N·m and an RMS torque = 0.103 N·m. The selected NEMA 17 Stepper Motor (Model: 17HS19-2004S1) has a maximum torque = 0.59 N·m (according to its datasheet). Since this value is much higher than the required peak torque = 0.115 N·m, the motor provides a large safety margin, ensuring reliable performance under acceleration and varying load conditions.

1.5 Actuator sizing for conveyor belt

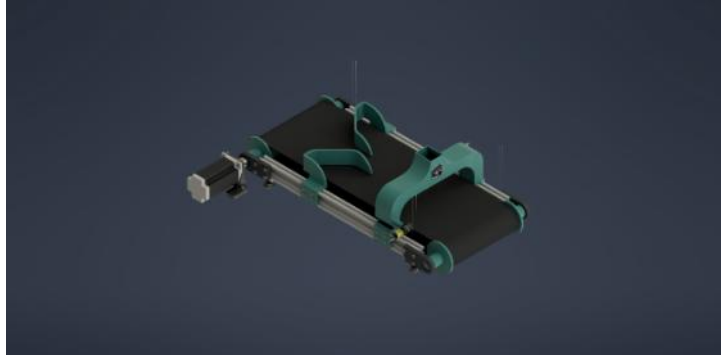


Figure 8: Conveyor Belt Mechanism

Symbol	Value	Unit
M_L	0.04	kg
M_C	0.20	kg
M_{total}	0.24	kg
r	0.02	m
L	0.13	m
ρ	1240	kg/m ³
η	0.45	
$S.f$	1.5	
$N_{rollers}$	2	
t_{acc}	0.70	s

Effective Load Inertia:

$$J_L = \frac{r^2 \times m_L}{\eta} = \frac{.04 \times (.02)^2}{.45} = 3.556 \times 10^{-5} \text{ kg.m}^2$$

Rollers Inertia:

$$J_{rollers} = \frac{\pi}{2} \rho L r^4 \times 2 = 8.103 \times 10^{-5} \text{ kg.m}^2$$

Equivalent Inertia:

$$J_{eq} = 2J_{Lt} = 7.87 \times 10^{-4} \text{ kg.m}^2$$

Load Force and Torque:

$$f_L = M_L \times g = .24 \times 9.8 = 2.352 \text{ N}$$

$$T_L = \frac{r \times f_L}{\eta} = 0.105 \text{ N.m}$$

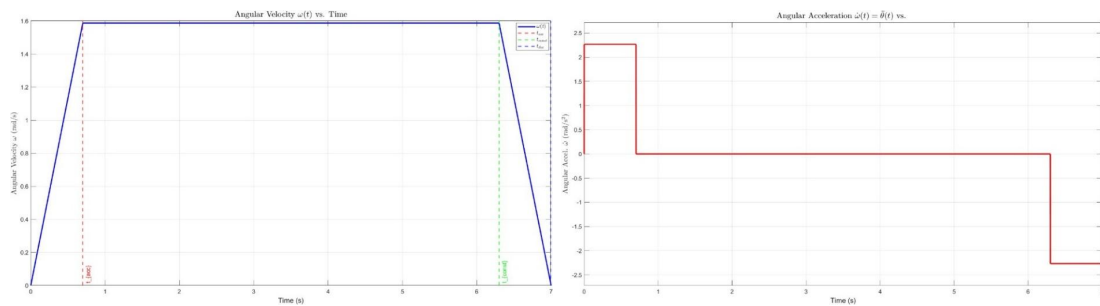


Figure 9: Velocity and Acceleration for Conveyor

Acceleration Torque:

$$Tm_{acc} = (7.87 \times 10^{-4} \times 2.27) + 0.105 = 0.1064 \text{ N.m}$$

$$T_{peak} = Tm_{acc} \times S.f = .16 \text{ N.m}$$

$$T_{rms} = \sqrt{\frac{0.1064^2 \times 0.7 + 0.105^2 \times 5.6 + 0.103^2 \times 0.7}{7}} \times 1.5 = 0.157 \text{ N.m}$$

5- Validation

Based on the analysis, the conveyor belt drive requires a peak torque = 0.16 N·m and an RMS torque = 0.157 N·m. The selected NEMA 17 Stepper Motor (Model: 17HS19-2004S1) has a maximum torque = 0.59 N·m, (as specified in its datasheet). Since this value is much higher than the required peak torque = 0.16 N·m, the motor provides a large safety margin, ensuring reliable performance during startup and under variable load conditions throughout continuous operation.

1.6 Actuator sizing for Feeder disk

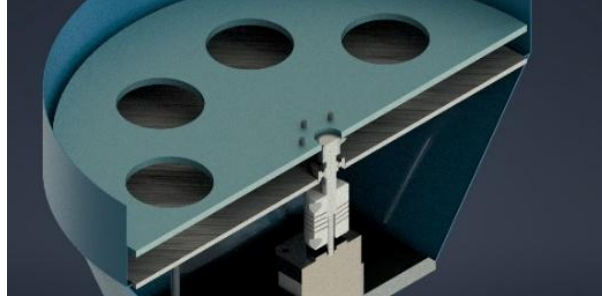


Figure 10: Feeder Disk

Symbol	Value	Unit
M_{disk}	0.15	kg
$M_{products}$	0.16	kg
M_{total}	0.31	kg
r_{disk}	0.02	m
d	0.1	m
I_{CG}	4.019	kg-mm ²

Disk Inertia:

$$J_{disk} = \frac{1}{2} M_{disk} \times r_{disk}^2 = .001172 \text{ kg.m}^2$$

Product Inertia:

$$J_{products} = I_{CG} + M_{products} \times d^2 = 4.019 \times 10^{-6} + .16 \times .1^2 = .001604 \text{ kg.m}^2$$

Equivalent Inertia:

$$J_{eq} = 2J_{Lt} = .005552 \text{ kg.m}^2$$

Load Torque:

$$f_L = (M_{disk} + M_{products}) \times g = .31 \times 9.8 = 3.04 \text{ N}$$

$$T_L = f_L \times d = 0.304 \text{ N.m}$$

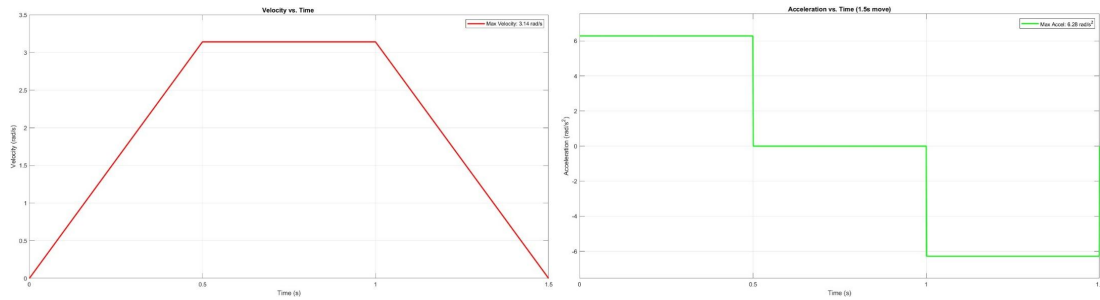


Figure 11: Feeder Disk Kinematics

$$Tm_{acc} = (.005552 \times 6.283) + 0.304 = 0.339 \text{ N.m}$$

$$T_{peak} = Tm_{acc} \times S.f = .51 \text{ N.m}$$

$$T_{rms} = \sqrt{\frac{0.339^2 \times 0.5 + 0.304^2 \times 0.5 + 0.27^2 \times 0.5}{1.5}} \times 1.5 = 0.46 \text{ N.m}$$

5- Validation

Based on the analysis, the feeding mechanism requires a peak torque = 0.51 N·m and an RMS torque = 0.46 N·m. The selected NEMA 17 Stepper Motor (Model: 17HS19-2004S1) has a maximum torque = 0.59 N·m, (as stated in its datasheet). Since this value is higher than the required peak torque = 0.51 N·m, the motor provides a sufficient safety margin, ensuring reliable and consistent performance for the feeding mechanism's operation

2 Proper Material Selection and Manufacturing Processes

Part Name	Category	Material	Manufacturing Type	Quantity
Feeder Disk	Feeder	PLA	3D Printing	1
Feeder Complementary Part	Feeder	PLA	3D Printing	1
Feeder Stepper Motor	Feeder	NEMA 17	Purchased	1
Conveyor Belt	Conveyor	Rubber	Purchased	1
Conveyor Complementary Parts	Conveyor	PLA	3D Printing	3
Conveyor Stepper Motor	Conveyor	NEMA 23	Purchased	1
Conveyor Frame/Rods	Conveyor	Aluminium	Purchased	2
Conveyor Rollers	Conveyor	PLA	3D Printing	2
X-axis Ball Screw	Robotic Arm	Carbon Steel	Purchased	1
Y-axis Ball Screw	Robotic Arm	Carbon Steel	Purchased	1
Z-axis Ball Screw	Robotic Arm	Carbon Steel	Purchased	1
Linear Rail Shafts	Robotic Arm	Stainless Steel	Purchased	6
Robot Gripper	Robotic Arm	Aluminium Alloy	Purchased	1
Gripper Servo Motor	Robotic Arm	Servo	Purchased	1
X-axis Stepper Motor	Robotic Arm	NEMA 17	Purchased	1
Y-axis Stepper Motor	Robotic Arm	DC Encoder	Purchased	1
Z-axis Stepper Motor	Robotic Arm	NEMA 17	Purchased	1
Wooden Base	Structural	Plywood	Carpentry	1
Motor Holders	Mechanical	Aluminium	Purchased	5
Product Lids (PLA)	Products	PLA	3D Printing	8
Product Bases (PLA)	Products	PLA	3D Printing	8
Control Box	Electrical	PLA	3D Printing	1

3 Stress analysis

3.1 Stress analysis Z-Axis

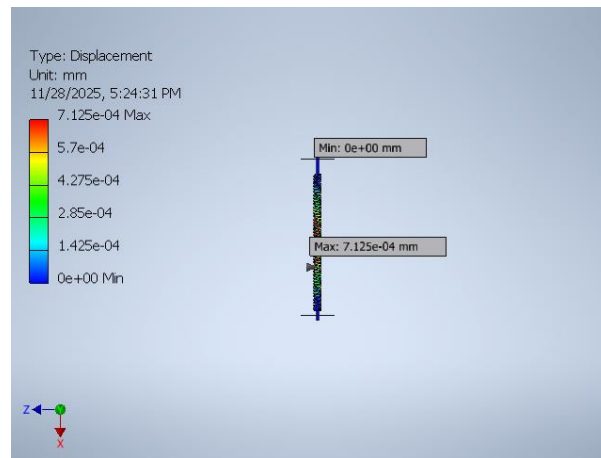


Figure 12: Stress analysis Z-Axis

Name	Minimum	Maximum
Von Mises Stress	0.0000237504 MPa	3.65433 MPa
1st Principal Stress	-0.0990244 MPa	2.52632 MPa
Displacement	0 mm	0.000712513 mm
Safety Factor	15 ul	15 ul

3.2 Stress analysis Y-Axis



Figure 13: Stress analysis Y-Axis

Name	Minimum	Maximum
Von Mises Stress	0.00000675196 MPa	98.5061 MPa
Displacement	0 mm	0.75984 mm
Safety Factor	7.69495 ul	15 ul

3.3 Stress analysis for X-Axis

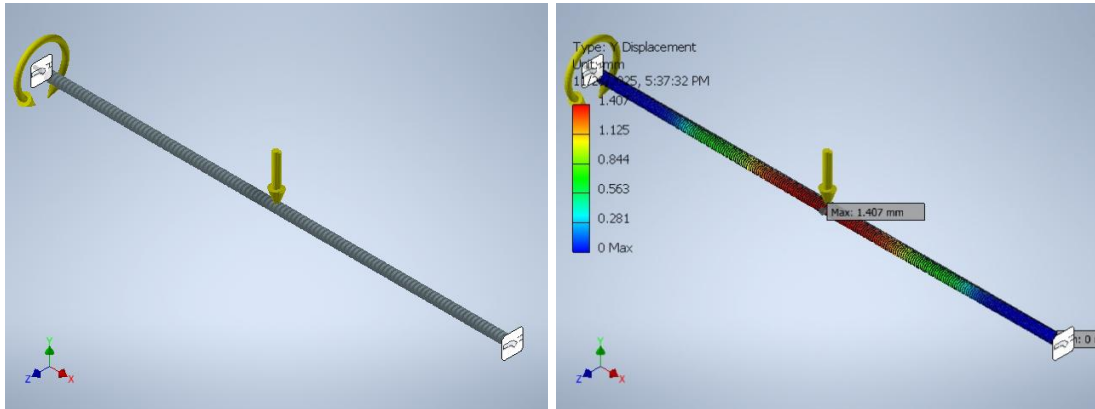


Figure 14: Boundary condition and Stress analysis for X-Axis

3.4 Stress analysis for conveyor belt

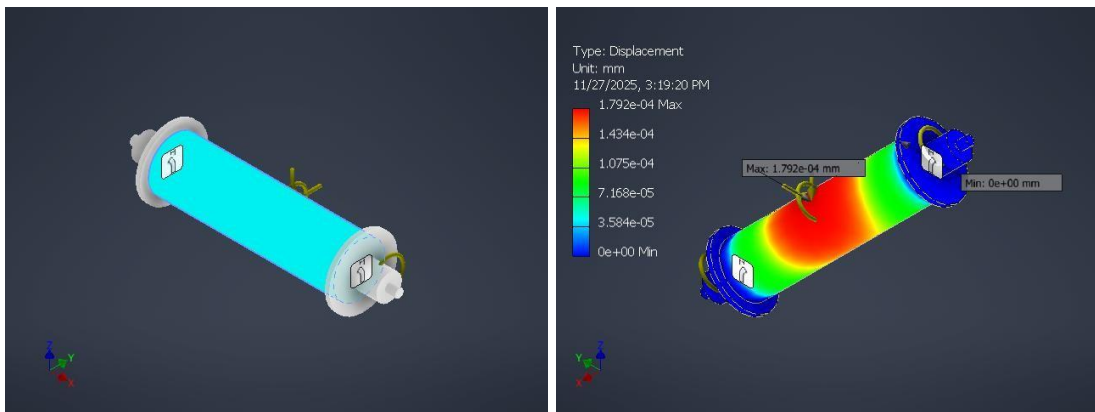


Figure 15: Boundary condition and Stress analysis for Conveyor Belt

Name	Minimum	Maximum
Von Mises Stress	0.000000154642 MPa	0.00790979 MPa
Displacement	0 mm	0.000179212 mm
Safety Factor	15 ul	15 ul

3.5 Stress analysis for Feeder disk

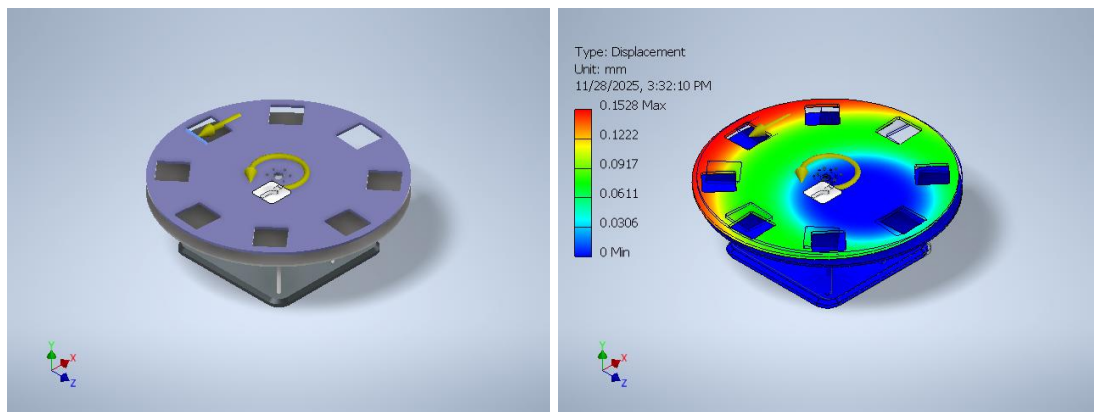


Figure 16: Boundary condition and Stress analysis for Feeder Disk

3.6 Stress analysis for Upper bracket

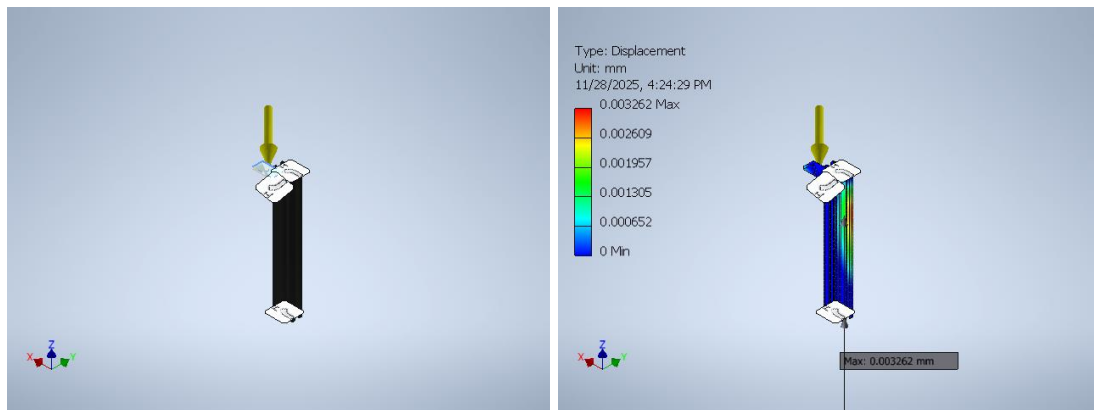


Figure 17: Boundary condition and Stress analysis for Upper Bracket

3.7 Stress analysis for Lower bracket

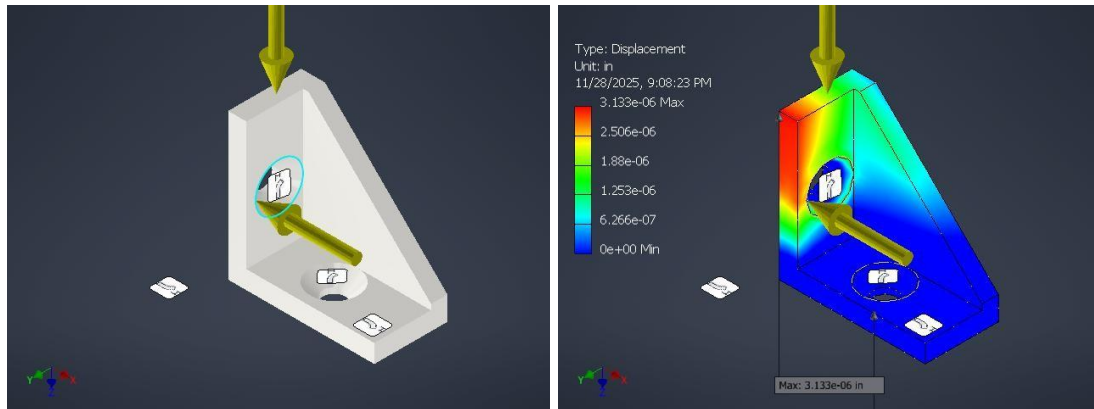


Figure 18: Boundary condition and Stress analysis for Lower bracket

Name	Minimum	Maximum
Von Mises Stress	0.00000000255795 ksi	0.92337 ksi
Displacement	0 in	0.00000318725 in
Safety Factor	15 ul	15 ul

3.8 Stress analysis for screw

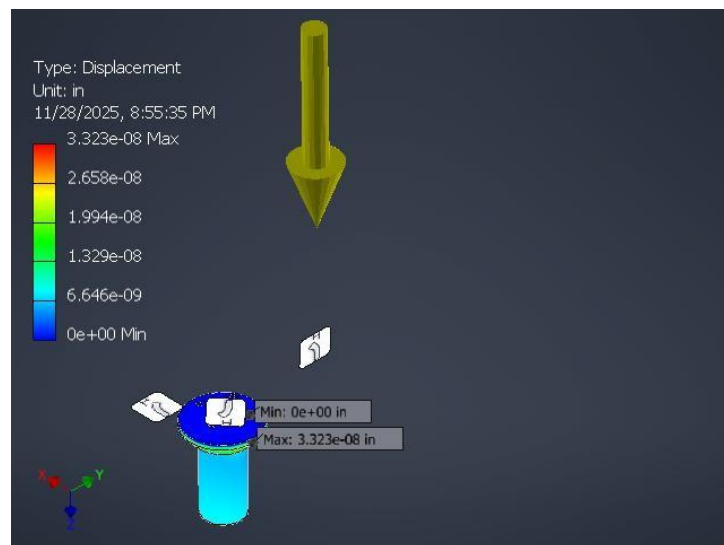


Figure 19: Stress analysis for screw